Mat2002 Tutorial 5

Junwei Zha

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Homogeneous Equation with Constant Coefficient I

Consider homogeneous equation with constant coefficients:

$$L(y) = ay'' + by' + cy = 0, \quad a, b, c \text{ are constants, } a \neq 0$$

Trial:

$$y = e^{rt}$$

$$y' = re^{rt}, \quad y'' = r^2 e^{rt}$$

Then $y' = re^{rt}, y'' = r^2 e^{rt}$ Substituted into ODE,
$\left(ar^2 + br + c\right) e^{rt} = 0$

$$\Leftrightarrow \quad ar^2 + br + c = 0. \text{ called characteristic equation.}$$

Solve the chacteristic equation, 3 cases:
(1) $\Delta = b^2 - 4ac > 0$, two different real roots, $r_1 \neq r_2$.
(2) $\Delta = b^2 - 4ac = 0$, repeated real root,$r_1 = r_2$.
(3) $\Delta = b^2 - 4ac < 0$, two conjugate complex roots, $\lambda \pm i\mu$.

Homogeneous Equation with Constant Coefficient II

The general solutions can be summarized as follows:

Discriminant	Nature of roots	General solutions
$b^2 - 4ac > 0$	$r_1 \neq r_2$, real	$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$
$b^2 - 4ac = 0$	$r_1 = r_2$, real	$y = c_1 e^{r_1 t} + c_2 t e^{r_1 t}$
$b^2 - 4ac < 0$	$r_1 = \lambda + i\mu, r_2 = \lambda - i\mu$	$y = c_1 e^{\lambda t} \cos \mu t + c_2 e^{\lambda t} \sin \mu t$

Example: Distinct Real Roots

Example

$$\begin{cases} y'' + 5y' + 6y = 0, \\ y(0) = 2, \quad y'(0) = 3. \end{cases}$$

Solution: Characteristic equation: $r^2 + 5r + 6 = 0$, roots $r_1 = -2$, $r_2 = -3$.

$$y = c_1 e^{-2t} + c_2 e^{-3t}$$

Applying initial conditions:

$$\begin{cases} c_1 + c_2 = 2, \\ -2c_1 - 3c_2 = 3, \end{cases} \Rightarrow c_1 = 9, \quad c_2 = -7$$

Final solution:

$$y = 9e^{-2t} - 7e^{-3t}$$

Example: Conjugate Complex Roots

Example

$$\begin{cases} y'' + y' + y = 0, \\ y(0) = 1, \quad y'(0) = 0. \end{cases}$$

Solution: Characteristic equation: $r^2 + r + 1 = 0$, roots $r = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$.

$$y = e^{-\frac{1}{2}t}\left(c_1 \cos\frac{\sqrt{3}}{2}t + c_2 \sin\frac{\sqrt{3}}{2}t\right)$$

Applying initial conditions: $\begin{cases} c_1 = 1, \\ -\frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 = 0, \end{cases} \Rightarrow c_1 = 1, \quad c_2 = \frac{1}{\sqrt{3}}$

Final solution:

$$y = e^{-\frac{1}{2}t}\left(\cos\frac{\sqrt{3}}{2}t + \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right)$$

Example: Repeated Roots

Example

$$y'' + 4y' + 4y = 0$$

Solution: Characteristic equation: $r^2 + 4r + 4 = 0$, repeated root $r = -2$.

$$y = c_1 e^{-2t} + c_2 t e^{-2t}$$

Euler Equation in Standard Form

Euler-Cauchy Equation

A linear differential equation of the form:

$$x^2y'' + Axy' + By = 0, x > 0$$

where A and B are constants.

Method of Solution

Key Idea

Look for solutions of the form $y = x^r$ (for $x > 0$)

1. Assume $y = x^r$
2. Compute derivatives:

$$y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}$$

3. Substitute into $x^2y'' + Axy' + By = 0$:

Characteristic Equation

After substitution:

$$x^2 \cdot r(r-1)x^{r-2} + Ax \cdot rx^{r-1} + Bx^r = 0$$

$$x^r[r(r-1) + Ar + B] = 0$$

Since $x^r \neq 0$ (for $x \neq 0$), we get:

Characteristic Equation

$$r^2 + (A-1)r + B = 0$$

Case 1: Distinct Real Roots

If $r_1 \neq r_2$ are real

General solution:

$$y = C_1 x^{r_1} + C_2 x^{r_2}$$

Example

Solve: $x^2 y'' + 2xy' - 2y = 0$

Characteristic: $r(r-1) + 2r - 2 = r^2 + r - 2 = 0$

Roots: $r_1 = 1, r_2 = -2$

Solution: $y = C_1 x + C_2 x^{-2}$

Case 2: Repeated Real Roots

If $r_1 = r_2 = r$

General solution:

$$y = (C_1 + C_2 \ln(x))x^r$$

Example

Solve: $x^2y'' - 3xy' + 4y = 0$
Characteristic: $r(r-1) - 3r + 4 = r^2 - 4r + 4 = 0$
Roots: $r = 2$ (double)
Solution: $y = (C_1 + C_2 \ln(x))x^2$

Case 3: Complex Conjugate Roots

If $r = \alpha \pm i\beta$

General solution:

$$y = x^{\alpha}[C_1 \cos(\beta \ln(x)) + C_2 \sin(\beta \ln(x))]$$

Example

Solve: $x^2y'' + 3xy' + 2y = 0$
Characteristic: $r(r-1) + 3r + 2 = r^2 + 2r + 2 = 0$
Roots: $r = -1 \pm i$
Solution: $y = x^{-1}[C_1 \cos(\ln(x)) + C_2 \sin(\ln(x))]$

Initial Value Problem

Problem

Solve: $x^2y'' - 2y = 0$ *with* $y(1) = 1, y'(1) = 0$

Initial Value Problem

Problem

Solve: $x^2y'' - 2y = 0$ *with* $y(1) = 1, y'(1) = 0$

Solution

1. *Characteristic:* $r(r-1) - 2 = r^2 - r - 2 = 0$
2. *Roots:* $r_1 = 2, r_2 = -1$
3. *General:* $y = C_1x^2 + C_2x^{-1}$
4. *Apply conditions:*

$$y(1) = C_1 + C_2 = 1$$
$$y'(1) = 2C_1 - C_2 = 0$$

5. *Solve:* $C_1 = \frac{1}{3}, C_2 = \frac{2}{3}$

Answer: $y = \frac{1}{3}x^2 + \frac{2}{3}x^{-1}$